Math 330 Abstract Algebra Notes

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1 Introduction to Group Theory

1.1 Modular Arithmetic

Definition 1.1. Take $n > 0$ and $n \in \mathbb{Z}$. We say that $a, b \in \mathbb{Z}$ are *congruent modulo* n ($a \equiv b \mod n$) if n divides $a - b$ ($n|a - b$) or if the remainder of n/a equals the remainder of n/b.

Here are some properties:

1. if $a \equiv b \mod n$ and $c \equiv d \mod n$, then $a + c \equiv b + d \mod n$.

   Proof. $a \equiv b \mod n$ implies that $n|a - b$ which implies that $\exists q \in \mathbb{Z}$ such that $a - b = nq$. The same can be said for $a \equiv b \mod n$ for some $q' \in \mathbb{Z}$ such that $c - d = nq'$. Adding these equations together we have that $a - b + c - d = (a + c) - (b + d) = n(q + q')$. Since the integers are closed under addition, $q + q'$ is an integer and therefore $n|(a + c) - (b + d)$ and as a result, $a + c \equiv b + d \mod n$. □

2. if $a \equiv b \mod n$ and $c \equiv d \mod n$, then $ac \equiv bd \mod n$

   Proof. $a \equiv b \mod n$ implies that $n|a - b$ which implies that $\exists q \in \mathbb{Z}$ such that $a - b = nq$. The same can be said for $a \equiv b \mod n$ for some $q' \in \mathbb{Z}$ such that $c - d = nq'$. We can write ac as

$$\begin{aligned} ac &= (b + nq)(d + nq') \\ &= bd + bnq' + dnq + n^2qq' \\ \Rightarrow ac - bd &= n(bq' + dq + nqq') \end{aligned}$$

and since everything in the parentheses are integers and the integers are closed under multiplication and addition, their result is an integer and therefore $n|ac-bd$ and $ac \equiv bd \mod n$. □

3. Congruence mod n is an equivalence relation

Proof. An equivalence relation is reflexive, symmetric, and transitive. To prove reflexiveness, we show $a \equiv a \mod n \to n|a-a \to n|0$ which is always true. To prove symmetry, we have that $a \equiv b \mod n$ implies $(a-b) = nq$ which we can extract a negative sign such that $(b-a) = n(-q)$ and since $-q$ is also an integer, $n|b-a$ and therefore $b \equiv a \mod n$. To prove transitiveness, we say that $a \equiv b \mod n$ implies $a-b = nq$ and $b \equiv c \mod n$ implies $b-c = nr$. We now find that $a-c = b+nq+nr-b = n(q+r)$. Since the integers are closed under addition, $q+r \in \mathbb{Z}$ and therefore $n|a-c$ and $a \equiv c \mod n$. □

1.2 Introduction to Groups

Definition 1.2 (Group)**.** A group is a set of elements G with a binary operation $*$ that assigns to every pair (g,h) of elements of G another element in G denoted $g*h$. Groups follow the group axioms:

1. Associativity
$$(g_1 * g_2) * g_3 = g_1 * (g_2 * g_3)$$
2. There exists an element $e \in G$ that is the identity such that
$$e * g_1 = g_1 * e = g_1$$
3. Every element has an inverse element denoted g^{-1} such that
$$g * g^{-1} = g^{-1} * g = e$$
4. The set of elements is closed under the group operation (although this is implied in the definition).

Example 1.3. Let us look at some examples

1. The set $(\mathbb{Z}, +)$. Addition is associative, the identity is 0, the inverse of each element g is $-g$.
2. The set $(\mathbb{R}_{>0}, \cdot)$ where $\cdot$ denotes regular multiplication. Regular multiplication is associative, the identity is 1, the inverse element is $\frac{1}{g}$
3. The set $(\{-2,-1,0,1,2\}, +)$. This fails due since it is not closed (take $1+2=3 \notin$ the set
4. The set $\left(\left\{\frac{1}{2^n}\right\}, \cdot\right)$. This fails since there is no inverse.
5. Fix a positive integer n. Take the set $(\{0,1,2,\ldots,n-1\}, + \mod n)$. The identity is 0, addition is associative. The inverse of an element $0 \le a \le n-1$ is $n-a$ since $(n-a)+a \mod n = n \mod n = 0$. This group is denoted by Fraleigh as $\mathbb{Z}_n$ but more generally denoted as $\mathbb{Z}/n\mathbb{Z}$.
6. Fix an $n \in \mathbb{N}$. We denote the group $\mathrm{GL}_n(\mathbb{R})$ as the set of invertible $n \times n$ matrices with real entries with the operation of matrix multiplication. Matrix multiplication is associative, the identity is the $n \times n$ identity matrix I_n. The inverse of every element A is the inverse matrix of A. This group is also known as the *General Linear Group*.

1.3 Properties of Groups

Definition 1.4. The *order* of a group G, denoted $|G|$,is the cardinality of the group/set. If G is finite, then $|G| < \infty$.

Definition 1.5. If the group operation $*$ is commutative, then the group is *Abelian*.

Here are some properties:

1. The identity is unique

Proof. Suppose we have some $f \in G$ such that $f * g = g \ \forall g \in G$. Then if we compose f with the identity, we should get $f * e = e$. By axiom 2 of the group axioms, we must also have that $f * e = f$ and therefore we have that $f * e = e = f$. □

2. Each inverse element is unique.

   Proof. $(a')'a' = e \to (a')'a'a = ea \to (a')'e = a \to (a')' = a$ □

3. Cancellation, that is if $a * b = c$ for some $a, b, c \in G$, then $b = a^{-1} * c$.

   Proof. By group axiom 3, there exists an inverse a^{-1}. We can then left multiply by a^{-1} such that $a^{-1} * a * b = a^{-1} * c$ and by associativity, we then have $b = a^{-1} * c$. □

   The consequences of this is that we can make the following observations: $a * b = a * c$ implies that $b = c$ and $b * a = c * a$ implies $b = c$.

For finite groups, we can write a group table where we have the left argument along the rows and the right argument of the operation along the columns. Since Abelian groups are commutative, they have a symmetry across the diagonal. An interesting thing to note with these group tables is that every entry appears only once in each row and column.

Proof. If we have a repeated entry in a row, that implies $a * b = a * c$ but by cancellation, we have that $b = c$. For each column, this implies that $b * a = c * a$ which also implies $b = c$. □

Example 1.6. Let us take a group of cardinality 2. We require that there exists an identity element so the set contains $\{e, a\}$ where a is some arbitrary element. We can immediately fill out the sides adjacent to the header rows by the group axioms (since $e * g = g$) in Table 1. Therefore we must have that $a * a = e$ and therefore a is its own inverse. We shall also note if we apply the group operation n-times to an element, we can represent it as $a^n = a * a * \ldots * a$ (n times). If we take a group of cardinality three, we have that $G = \{e, a, b\}$. We once again

	e	a
e	e	a
a	a	e

Table 1: Group table for a group of cardinality 2.

can fill out the header adjacent row and column by composition with the identity. In the $a * b$ cell of Table 2, we require that be e because otherwise, that would require $a * a$ to be e and therefore $b * a = a$ but a was already in that column and so hence we have a contradiction. Therefore we must have $a * a = b$ and $a * b = e$. From this, we can see that $a^2 = b$ and if we multiply $a * b = e$, this implies $a * a^2 = a^3 = e$. Our group is therefore $G = \{e, a, a^2\}$.

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

Table 2: Group table for a group of cardinality 3

Example 1.7. Let us look at the Quaternion group. The Quaternion group is the set

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$$

under standard multiplication. 1 is the identity element and $i^2 = j^2 = k^2 = -1$. Quaternion multiplication is defined as

$$\begin{aligned} ij = k \qquad jk = i \qquad ki = j \\ ji = -k \qquad kj = -i \qquad ik = -j \end{aligned}$$

This is a group of order 8 and is non-abelian. The inverse of -1 is -1 and so because of this we can say the inverse i and $-i$ and same for j and k. Associativity is also good. This group is the reason why the cross product exists in $\mathbb{R}^3$ and the related Octonion group allows for the existence of the cross product in $\mathbb{R}^7$.

Example 1.8. Another important group is the set of complex numbers with norm of 1 under multiplication. In other words, $\{z \in \mathbb{C} : |z| = 1\}$. This is an Abelian group and is of infinite order. This is a group because the norms are all one so multiplication stays within the group ($|z_1||z_2| = |z_1 z_2| = 1$) and the inverse is also in the group (if $|z_1| = 1, \frac{1}{|z_1|} = 1$). We can also convert each complex number into its polar form $z = re^{i\theta}$ $r, \theta \in \mathbb{R}$ and since each element has a norm of 1, $r = 1$ and we can just take $z_1 z_2 = e^{i(\theta_1+\theta_2)}$ and since the real numbers are a group under addition, this set of complex numbers also form a group. This group is also used to generate groups which contain the nth primitive roots of unity. To do this, we fix a number $n \in \mathbb{N}$. Then the number

$$e^{\frac{2k\pi}{n}i} \qquad k \in \{1, 2, \ldots, n-1\}$$

is an nth primitive root of unity. This means that these are roots to the equation $z^n = 1$ and the primitive means that n is the smallest power that can satisfy the equation (e.g., no smaller number (let's call m) exists such that $z^m = 1$). We then say that the set of the nth groups of unity form an Abelian group under multiplication of order n. One can see this because multiplication of complex numbers is associative, the identity is 1 ($k = 0$ root)

Definition 1.9. Suppose A is any set. A permutation is a function $f : A \mapsto A$ that is bijective.

Proposition 1.10. *The set of all permutations of a set A is a group where the binary operation $*$ is composition and e is the identity permutation.*

Proof. We must first show that the composition of permutations is also a permutation. Let us say that σ, τ are permutations of A. We can now prove that $\sigma\tau$ is injective. Suppose

$$(\sigma\tau)(a) = (\sigma\tau)(b) \to \sigma(\tau(a)) = \sigma(\tau(b))$$

Since σ is a permutation, σ is injective and therefore

$$\tau(a) = \tau(b)$$

Using that logic again, τ is a permutation and therefore injective and

$$a = b$$

To show that this is onto, we have to show that for some $a \in A$, there exists an $x \in A$ such that $a = \sigma\tau(x)$ Since σ is onto, there exists a $y \in A$ such that $a = \sigma(y)$. τ is also surjective and so we can say there exists an $x \in A$ such that $y = \tau(x)$. Putting everything together we have that

$$a = \sigma(y) = \sigma(\tau(x)) = (\sigma\tau)(x)$$

As such we have shown that $\sigma\tau$ is a permutation. Now we can check the group axioms. The group operation is associative since function is associative

$$((\sigma\tau)\rho)(a) = \sigma(\tau(\rho(a))) = \sigma(\tau\rho(a)) = (\sigma(\tau\rho))(a)$$

We also denote the identity permutation ι such that $\iota(a) = a \ \ \forall a \in A$. Finally, any permutation is invertible since it surjective and given any $y \in A$, there exists only one x such that $y = \sigma(x)$ since σ is bijective. We therefore have that $\sigma^{-1}(y) = x$. We can check that σ^{-1} is a permutation by using the fact that $\sigma(\sigma^{-1}(x)) = x$. Let us consider $\sigma^{-1}(y) = \sigma^{-1}(x)$ and applying σ to both sides we get $x = y$ and hence injectivity. We shall define a $y = \sigma(x)$ and therefore $\sigma^{-1}(y) = \sigma^{-1}(\sigma(x)) = x$ for all $y \in A$. □

This group is known as the symmetric group and we denote it S_A. The most important symmetric group is when $A = \{1, 2, \ldots, n\}$. We denote this group as S_n and it has cardinality $|S_n| = n!$.

From now on, we will write $a * b$ as ab except when working with groups using $\mathbb{Z}, \mathbb{R}$, etc. with addition as they will be denoted using the usual addition sign. Instead of denoting the identity as e, we shall write 1 under the same condition as the last where there is no confusion is possible.

1.4 The Symmetric Group

Recall that the Symmetric Group S_n is the defined as the set of all permutations of a set $\{1, 2, \ldots, n\}$ where the binary operation is composition. The cardinality of S_n is $|S_n| = n!$. Elements of S_n can be written as

$$\begin{pmatrix} 1 & 2 & 3 & 4 & \ldots & n \\ k_1 & k_2 & k_3 & k_4 & \ldots & k_n \end{pmatrix} \text{ where } k_1, \ldots, k_n \in \{1, \ldots, n\}$$

Since we are using composition, we do the permutation τ before σ in the permutation $\sigma\tau$. Let us look at an example to demonstrate

Example 1.11. Let us suppose in S_5

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \end{pmatrix} \qquad \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 2 & 5 & 4 \end{pmatrix}$$

The permutation $\sigma\tau$ is then

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 4 & 3 \end{pmatrix}$$

We can also find the inverse of a permutation by just going in "reverse"

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 5 & 1 \end{pmatrix}$$

Proposition 1.12. *If $n \geq 3$, the group S_n is non-Abelian*

Proof. Let us first just consider the case $n = 3$. Consider the following composition of permutations

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

We can prove this for higher n by simply sending everything beyond 3 to their number. For example

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & \ldots \\ 2 & 3 & 1 & 4 & 5 & \ldots \end{pmatrix}$$

□

1.4.1 Cycle Notation

We can express the elements of a group in a more streamlined way.

$$(k_1, k_2, \ldots, k_l) \text{ where } k_1, \ldots, k_l \in \{1, 2, \ldots, n\} \text{ and are distinct}$$

This notation denotes the following permutation: $k_1 \to k_2 \to \ldots \to k_{l-1} \to k_l \to k_1$. Any numbers not listed in this cycle is assumed to be fixed. In this case we have to specify what symmetric group we are in because the number of fixed entries could be 1 or 2 or 500.

Example 1.13. In S_6, the notation $(3, 1, 5, 4)$ represents the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 1 & 3 & 4 & 6 \end{pmatrix}$$

We can also compose them together as normal

$$(3, 1, 5, 4)(1, 2, 6, 5, 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 5 & 3 & 1 & 4 \end{pmatrix}$$

If we want to invert a permutation, we just reverse the order of the terms. For example $\sigma = (k_1, k_2, \ldots, k_l)$ inverts to $\sigma^{-1} = (k_l, k_{l-1}, \ldots, k_1)$. Cycles of length 2 are specifically called *transpositions*. We also note that cycles are not unique since we see that the inverse of a cycle (a, b) is (b, a) which is the same as (a, b). There is a fact that if you have 2 cycles are disjoint, that is, they don't have any entries in common, the disjoint cycles commute. Additionally, every permutation can be written as a product of disjoint cycles.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 5 & 3 & 6 & 7 & 4 & 2 & 1 \end{pmatrix} = (1, 8)(2, 5, 7)(3)(4, 6)$$

The cycles with just a single element can be excluded.

1.5 The Dihedral Group

Start with a regular n-gon. The dihedral group D_n is the group of all symmetries of this n-gon. Let us take a look at the symmetries of a triangle.

Example 1.14. We can consider the symmetries of triangle as a clockwise rotation by 120° once or twice (which is the same as a counter-clockwise rotation by 120°) or reflecting along an axis of symmetry that goes through one of the vertices. These are described in Figure 1. We can call one clockwise rotation by ρ_1, two clockwise (or again,

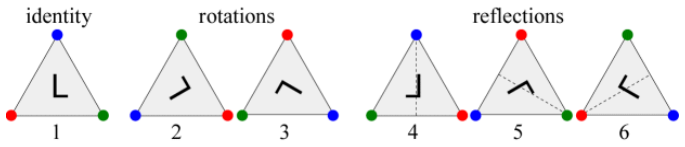


Figure 1: Elements of the Dihedral Group D_3

1 counter-clockwise rotation) by ρ_2, and then we have reflections μ_1, μ_2, μ_3 about the vertices $1, 2$, and 3. We also note that $\rho_1^2 = \rho_2 = 1$ and $\mu_1^2 = \mu_2^2 = \mu_3^2 = 1$. We claim that $D_3 = \{1, \rho_1, \rho_2, \mu_1, \mu_2, \mu_3\}$ are all the elements of the dihedral group D_3. What about products though? We can think of dihedral group as a subset of the symmetric group and so the symmetries of the n-gon are just permutations of the vertices. However, we need to preserve the relationships between the vertices (basically, there must exist an edge between 1 and 2 and symmetries that don't respect this edge, aren't part of the dihedral group). As such, we can write ρ_1 and μ_2 as

$$\rho_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \qquad \mu_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

We can then find the products of these two actions

$$\rho_1\mu_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = \mu_1 \qquad \mu_2\rho_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = \mu_3$$

We see that the product of any two symmetries is also a symmetry. We also see the D_n is non-Abelian (which makes sense because we established previously in Proposition 1.12).

Proposition 1.15. *The cardinality of the dihedral group D_n is $|D_n| = 2n$.*

Proof. We can express any rotation as sending the vertex 1 to any vertex $i \in \{1, \ldots, n\}$. Vertex 2 can then only be sent to $i-1$ or $i+1$ (which come about from reflection). Once these two are established, the rest of the vertices are completely determined due to the "rigid" edge requirement.

Let ρ be a single rotation clockwise by an angle $\frac{2\pi}{n}$, then there are n distinct rotations $\{1, \rho, \rho^2, \ldots, \rho^{n-1}\}$ (where $\rho^n = 1$). We can let μ be a reflection about an axis containing the center and vertex 1. We can then achieve any reflection by rotating a certain amount of times between 0 and $n-1$and then reflection. Our Dihedral group can then generalized to be $D_n = \{1, \rho, \ldots, \rho^{n-1}, \mu, \mu\rho, \ldots, \mu\rho^{n-1}\}$. Since there are n number of rotations and n vertices we can reflect about, we have $2n$ elements in our group.

We must also check that all elements are distinct. Each rotation is distinct since if not, then there exists two numbers $i, j \in \{0, \ldots, n\}$ such that $\rho^i = \rho^j \to \rho^{i-j} = e$ which isn't true unless $i = n$ and $j = 0$ since that is the only way to get back the identity permutation. For the reflection permutations, this would imply $\mu\rho^i = \mu\rho^j \to \rho^i = \rho^j$ for some $i, j \in \{0, \ldots, n\}$ which we established isn't possible. □

Proposition 1.16. $\rho\mu = \mu\rho^{-1}$

Proof.

$$\rho = \begin{pmatrix} 1 & 2 & 3 & \ldots & n-2 & n-1 & n \\ 2 & 3 & 4 & \ldots & n-1 & n & 1 \end{pmatrix} \qquad \rho^{-1} = \begin{pmatrix} 1 & 2 & 3 & \ldots & n-2 & n-1 & n \\ n & 1 & 2 & \ldots & n-3 & n-2 & n-1 \end{pmatrix}$$

$$\mu = \begin{pmatrix} 1 & 2 & 3 & \ldots & n-2 & n-1 & n \\ 1 & n & n-1 & \ldots & 4 & 3 & 2 \end{pmatrix}$$

$$\rho\mu = \begin{pmatrix} 1 & 2 & 3 & \ldots & n-2 & n-1 & n \\ 2 & 1 & n & \ldots & 5 & 4 & 3 \end{pmatrix} \qquad \mu\rho^{-1} = \begin{pmatrix} 1 & 2 & 3 & \ldots & n-2 & n-1 & n \\ 2 & 1 & n & \ldots & 5 & 4 & 3 \end{pmatrix}$$

□

1.6 Subgroups

I am making this a subsection because Tate Darin (who can be contacted at tatedarin28@u.northwestern.edu) said it would be funny if subgroups were a subsection.

Definition 1.17. Suppose G is a group. If H is a subset of G which is also a group under the same group operation defined on G, then H is a *subgroup* of G. We denote this as $H \leq G$.

Being a subgroup implies the following

1. 1 (which is the identity in G) $\in H$
2. The group operation needs to be closed on H
3. Inverse closure is also required

Every group G has at least 2 subgroups: the trivial subgroup ($\{1\}$) and G itself. If we want a subgroup that is neither of these, we refer to them as *proper, nontrivial* subgroup.

Example 1.18. If we take $G = \mathbb{Z}$ (if we do not specify which group operation with $\mathbb{Z}$, it is assumed to be addition), then a subgroup would by a group $n\mathbb{Z}$ where $n\mathbb{Z}$ is the set of integers divisible by n. In fact, we can state a theorem about the subgroups of $\mathbb{Z}$. However, first we must prove the division algorithm.

Theorem 1.19 (Division Algorithm)**.** *Let $a, b \in \mathbb{Z}$ with $b > 0$. Then there exist unique integers q and r such that $a = bq + r$ with $0 \leq r < b$.*

Proof. Let $S = \{a - bt : t \in \mathbb{Z}, a - bt \geq 0\}$. If $0 \in S$, then $a - bq$ for some $q \in \mathbb{Z}$ and therefore $a = bq + 0$. We can also claim that $S \subseteq \mathbb{N}$. We claim that S is nonempty. Let $t = -|a|$, then $a - bt = a + |a|b$. If $a \geq 0$, then $a + |a|b \geq 0$ since $b > 0$. If $a < 0$ then $a + |a|b = a(1 - b)$. Since $a < 0$ and $b \geq 1$, we get that $(1 - b) \leq 0$ and generally $a(1 - b) \geq 0$.

Since S is nonempty we can apply the Well Ordering Principle and so S has a least element we call $r = a - bq$. By definition, $r \geq 0$ and if $r \geq b$, then $0 \leq r - b < r$ but also $r - b = a - bq - b = a - b(q + 1)$ and therefore $a - b(q + 1)$ is a smaller element of S than r which contradicts our choice of r. Thus we must have $a = bq + r$ such that $0 \leq r < b$.

To prove uniqueness, let us supposed $a = bq_1 + r_1 = bq_2 + r_2$ with $q_i, r_i \in \mathbb{Z}$ and $0 \leq r_i < b$. Then $b(q_1 - q_2) = r_2 - r_1$. Specifically we have $b|q_1 - q_2| = |r_2 - r_1|$. If $q_1 \neq q_2$, then $b|q_1 - q_2| \geq b$ but at the same time $0 \leq r_1, r_2 < b \rightarrow |r_2 - r_1| < b$ which is impossible. Therefore $q_1 = q_2$ and $r_1 = r_2$. □

Theorem 1.20. *Every subgroup of $\mathbb{Z}$ is of the form $n\mathbb{Z}$.*

Proof. Suppose $H \leq \mathbb{Z}$. If $H = \{0\}$, there is nothing show $n\mathbb{Z}$. Otherwise, H contains at least one positive number since if H contains a negative number, we need its inverse which is its negative which is positive. We now say n is the smallest positive number in H by the Well Ordering Principle. Since $n \in H$, $n\mathbb{Z} \subseteq H$ since we require that $n^2, n^3, \ldots$ and so $n\mathbb{Z}$. Let us suppose $a \in H$ and we say that $a = qn + r$ $0 \leq r < n$. Since $a \in H$ and n is also $\in H$, $r = a - qn$ so $r = a - qn \in H$ and $r < n$. But n is the smallest positive number in H, we must have that $r = 0$ and therefore $n|a \rightarrow a = nq \in n\mathbb{Z}$. Since a is arbitrary, this means that $H = n\mathbb{Z}$. □

Proposition 1.21. *If $H_i \leq G$ $\forall i$, then $\bigcap H_i \leq G$.*

Proof. $1 \in \bigcap H_i$ since $1 \in H_i$ $\forall i$. That proves the identity is in the intersection. If $x, y \in \bigcap H_i$, then $x, y \in H_i$ $\forall i$ so $xy \in H_i$ $\forall i$ (by the definition of a subgroup), so $xy \in \bigcap H_i$ which proves closure. Finally, $x \in \bigcap H_i$ so $x \in H_i$ $\forall i$, and by the definition of a subgroup, the inverse $x^{-1} \in H_i$ $\forall i$ so $x^{-1} \in \bigcap H_i$. □

1.7 Generators

Definition 1.22. Suppose S is a subset of G. We define the subgroup *generated* by S (denoted $\langle S \rangle$) to be the smallest subgroup of G that contains S. In other words,

$$\langle S \rangle = \bigcap_{\substack{H \leq G \\ S \subseteq H}} H$$

Example 1.23. In $\mathbb{Z}$, $\langle 6, 9 \rangle$ generates the subgroup $3\mathbb{Z}$. In $\mathbb{Z}_6$, $\langle 2 \rangle$ generates $\{0, 2, 4\}$.

Generators are generally not unique.

Example 1.24. $Z_8 = \langle 1\rangle = \langle 3\rangle = \langle 5\rangle = \langle 7\rangle$.

One observation is that if G is a group and $H \leq G$, then if $S \subseteq H$, we must have $\langle S\rangle \subseteq H$.

Definition 1.25. A group that is generated by one element is deemed *cyclic*.

Proposition 1.26. *Any cyclic is abelian*

Proof. Say $G = \langle x\rangle = \{x^n : n \in \mathbb{Z}\}$. Let us propose

$$x^n x^m = x^{n+m} = x^{m+n} = x^m x^n$$

by the commutativity of addition. □

Theorem 1.27. *Any subgroup of a cyclic group is itself cyclic.*

Proof. Suppose $G = \langle a\rangle$ and $H \leq G$. If H is trivial, then $\{e\}$ generates itself and there is nothing to show. Otherwise, let a^n be the smallest positive power of a in H. Since $a^n \in H$, we therefore have that $\langle a^n\rangle \subseteq H$. We now must show the reverse inclusion. Take any $a^k \in H$ for some k since $H \leq G$ and G is cyclic. We now use the division algorithm to say that $k = qn + r$ for $0 \leq r < n$. We then have $a^k = a^{qn+r} = (a^n)^q a^r$. $(a^n)^q \in H$ since it is a multiple of the smallest power of $a \in H$. Using cancellation, we have $a^r = (a^n)^{-q} a^k \in H$ and since the right hand side is closed under the group operation, we have that $a^r \in H$ but since a^n is the smallest power of $a \in H$, we must have that $r = 0$. Since a^k is some arbitrary element in H, we have that $H \subseteq \langle a^n\rangle$ which establishes that $H = \langle a^n\rangle$. □

Let us look at an example of the subgroups of $\mathbb{Z}_{12}$.

Example 1.28. $\{0\}$ is the trivial subgroup. We can see that $\langle 1\rangle = \mathbb{Z}_{12}$. $\langle 2\rangle = \{0, 2, 4, 6, 8, 10\}$. $\langle 3\rangle = \{0, 3, 6, 9\}$. $\langle 4\rangle = \{0, 4, 8\}$. $\langle 5\rangle = \mathbb{Z}_{12}$. $\langle 6\rangle = \{0, 6\}$. $\langle 7\rangle = \mathbb{Z}_{12}$. $\langle 8\rangle = \langle 4\rangle$. $\langle 9\rangle = \langle 3\rangle$. $\langle 2\rangle = \langle 10\rangle$. $\langle 11\rangle = \mathbb{Z}_{12}$.

This example is a demonstration of a theorem we will soon prove. Let us first state some definitions

Definition 1.29. If $a, b \in \mathbb{Z}\backslash 0$, their *greatest common divisor* is the unique $g \in \mathbb{Z}^+$ satisfying

1. $g|a$ and $g|b$
2. If $c|a$ and $c|b$ then $c|g$

Theorem 1.30 (Bezout's)**.** *Given any nonzero integers a, b we can find integers m, n such that $ma + nb = \gcd(a, b)$*

We can now state our theorem.

Theorem 1.31. *In Z_n, $\langle a\rangle = \langle \gcd(a, n)\rangle$. Particularly, generators of $\mathbb{Z}_n$ are precisely those that are relatively prime to n.*

Proof. Find k, l such that $ka + ln = \gcd(a, n)$. Then in $\mathbb{Z}_n$, we have that $ka \equiv \gcd(a, n)$. Thus we have that $\langle \gcd(a, n)\rangle \subseteq \langle a\rangle$. Conversely, a is a multiple of $\gcd(a, n)$ so $\langle a\rangle \subseteq \langle \gcd(a, n)\rangle$ so now $\langle a\rangle = \langle \gcd(a, n)\rangle$. □

2 Homomorphisms

2.1 Properties of Homomorphisms

Consider the following groups: $\mathbb{Z}_2 = \{0, 1\}$ under addition mod 2, $\{1, -1\}$ under multiplication, and

$$\left\{ I_2, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$$

All these groups are of the form $\{e, a\}$ where $a \neq e$ and $a^2 = e$. In other words, they all have the same group table (1). In some sense, all these groups are "the same."

Definition 2.1. A function $\varphi : G_1 \mapsto G_2$ is a *homomorphism* if it satisfies $\varphi(x * y) = \varphi(x) * \varphi(y)$ where the left hand side is the group operation on G_1 and the right hand side is the group operation on G_2.

Proposition 2.2. *If $\varphi : G_1 \mapsto G_2$ is a homomorphism, then*

1. $\varphi_1(e_1) = e_2$

2. $\varphi(g^{-1}) = \varphi(g)^{-1} \ \forall g \in G$

Proof. To prove the first fact, we have that $e_1e_1 = e_1$ so $\varphi(e_1e_1) = \varphi(e_1)$ but also by the definition of a homomorphism, $\varphi(e_1e_1) = \varphi(e_1)\varphi(e_1)$ which implies $\varphi(e_1)\varphi(e_1) = \varphi(e_1) \to \varphi(e_1) = e_2$. To prove the second fact, we know that $g^{-1}g = e_1$ and so we apply φ to get $\varphi(g^{-1}g) = \varphi(e_1)$ but also is equal to $\varphi(g^{-1})\varphi(g)$ and so therefore $\varphi(g^{-1})\varphi(g) = e_2 \to \varphi(g^{-1}) = \varphi(g)^{-1}$ □

Definition 2.3. Suppose $\varphi : G_1 \mapsto G_2$ is a homomorphism. We define the *kernel* of φ as the set $\ker\varphi = \{x \in G_1 : \varphi(x) = e_2\}$ and the *image* of φ as the set $\operatorname{im}\varphi = \{y \in G_2 : y = \varphi(x) \text{ for some } x \in G_1\} = \{\varphi(x) : x \in G_1\}$.

Proposition 2.4. $\ker\varphi$ *is a subgroup of its domain* G_1 *and* $\operatorname{im}\varphi$ *is a subgroup of the codomain* G_2*.*

Proof. To prove that the kernel of φ is a subgroup of G_1, we show that $\varphi(e_1) = e_2 \therefore e_1 \in \ker\varphi$. If $\varphi(x) = e_2 = \varphi(y)$, then $\varphi(xy) = \varphi(x)\varphi(y) = e_2e_2 = e_2$ and therefore it is closed. Finally, if $\varphi(x) = e_2$, then $\varphi(x^{-1}) = \varphi(x)^{-1} = e_2^{-1} = e_2$ and inverse closure exists in $\ker\varphi$. Hence, $\ker\varphi \leq G_1$.

Proving the image of φ is a subgroup of G_2 is much of the same. The identity is in the image since $\varphi(e_1) = e_2$. If $y_1, y_2 \in \operatorname{im}\varphi$, then $y_1y_2 = \varphi(x_1)\varphi(x_2) = \varphi(x_1x_2)$ and hence it is closed. By the previous proposition, we know that $y^{-1} = \varphi(x^{-1})$ and hence the inverse also exists in the image. Hence, $\operatorname{im}\varphi \leq G_2$. □

Let us look at an example.

Example 2.5. Let us define a mapping $\det : GL_2(\mathbb{R}) \to \mathbb{R}^\times$ where $\mathbb{R}^\times$ are nonzero real numbers under multiplication. The determinant is a homomorphism since $\det(AB) = \det A \det B$. The kernel of the determinant is the set of all 2×2 invertible matrices with determinant 1 and is represented by the subgroup $SL_2(\mathbb{R})$. The image of the determinant is just $\mathbb{R}^\times$ since if $a \in \mathbb{R}^\times$, it can be represented as

$$a = \det\begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}$$

There always exists two homomorphisms: the trivial and identity homomorphisms. The trivial homomorphism is defined as $\varphi : G_1 \mapsto G_2$ by $\varphi(x) = e_2$ where the kernel is $\ker\varphi = G_1$ and the image is $\operatorname{im}\varphi = \{e_2\}$. The identity homomorphism is defined as $\mathrm{id}_G : G \mapsto G$ such that $\mathrm{id}(x) = x$. The kernel is $\ker\mathrm{id}_G = \{e\}$ and the image is $\operatorname{im}\mathrm{id}_G = G$.

2.2 Isomorphisms

Definition 2.6. A homomorphism is an isomorphism if it is invertible. More precisely, $\varphi : G_1 \mapsto G_2$ is an isomorphism if it is a homomorphism and there exists another homomorphism $\varphi^{-1} : G_2 \mapsto G_1$ such that $\varphi^{-1} \circ \varphi = \mathrm{id}_{G_1}$ and $\varphi \circ \varphi^{-1} = \mathrm{id}_{G_2}$.

Two groups are isomorphic if there exists an isomorphism between them which is denoted by

$$G_1 \cong G_2$$

Theorem 2.7. *A homomorphism is an isomorphism if and only if it is bijective.*

Proof. ($\Rightarrow$) Suppose we have an isomorphism $\varphi : G_1 \mapsto G_2$ so therefore φ^{-1} exists. To prove φ is injective, we have that $\varphi(x_1) = \varphi(x_2) \to \varphi^{-1}\varphi(x) = \varphi^{-1}\varphi(x) \to x_1 = x_2$. To prove surjectivity, let $y \in G_2$. φ^{-1} exists so let $x = \varphi^{-1}(y) \in G_1$. $\varphi(x) = \varphi\varphi^{-1}(y) = y$ which proves surjectivity.

($\Leftarrow$) Now suppose φ is bijective. We now need to find φ^{-1}. Let φ^{-1} be as follows.; $y \in G_2$ and $\varphi^{-1}(y) =$ the unique $x \in G_1$ for which $y = \varphi(x)$. This exists since φ is surjective and is unique since φ is injective. We then have $\varphi \circ \varphi^{-1}(y) = \varphi(\varphi^{-1}(y)) = \varphi($ the unique $x \in G_1$ such that $\varphi(x) = y) = y$. Now we check the other arrangement $\varphi^{-1} \circ \varphi(x) = \varphi^{-1}(\varphi($unique x such that $\varphi(x) = y)) = \varphi^{-1}(y) =$ the unique x such that $\varphi(x) = y$. This is our inversion. We still need to prove it is a homomorphism. Suppose $y_1y_2 \in G_2$ such that $\varphi^{-1}(y_1y_2) =$ unique $x \in G_1$ such that $\varphi(x) = y_1y_2$. We know that $\varphi^{-1}(y_1) = x_1 \in G_1$ and $\varphi^{-1}(y_2) = x_2$. Since φ is a homomorphism, we have that $\varphi(x_1x_2) = \varphi(x_1)\varphi(x_2) = y_1y_2$ and therefore $\varphi^{-1}(y_1y_2) = x_1x_2 = \varphi^{-1}(y_1)\varphi^{-1}(y_2)$. We therefore have proved that φ^{-1} is a homomorphism. □

Proposition 2.8. *A homomorphism is injective if and only if* $\ker\varphi = \{e_1\}$.

Proof. ($\Rightarrow$) Suppose φ is injective. $\ker\varphi = \{x \in G_1 : \varphi(x) = e_2\}$. We have that $\varphi(e_1) = e_2$ and since φ is injective, if $\varphi(x) = e_2$, $x = e_1$ and $\ker\varphi = \{e_1\}$.

($\Leftarrow$) Now suppose $\ker\varphi = \{e_1\}$.If $\varphi(y) = \varphi(h)$ then $\varphi(y)\varphi(h)^{-1} = e_2 \to \varphi(gh^{-1}) = e_2$ since φ is a homomorphism. We therefore have that $gh^{-1} \in \ker\varphi$ but since we're assuming the only element in the kernel is e_1, we have that $gh^{-1} = e_1 \to g = h$ and therefore φ is injective. □

In a similar way, one can prove that φ is surjective if and only if $\operatorname{im}\varphi = G_2$.

An isomorphism effectively means that if you rename the elements, two groups have the same group table.

Proposition 2.9. *Any cyclic group of order n is isomorphic to $\mathbb{Z}_n$ and any cyclic group of order ∞ is isomorphic to $\mathbb{Z}$.*

Proof. Suppose G is cyclic and has order n, then $G = \{1, a, a^2, \ldots, a^{n-1}\}$ where $a^n = 1$. We now define $\varphi : G \to \mathbb{Z}_n$ by $\varphi(a^k) = k \mod n$. In G, we see that $a^k a^l = a^{(k+l) \mod n}$ so $\varphi(a^k a^l) = (k+l) \mod n = \varphi(a^k) + \varphi(a^l)$ in $\mathbb{Z}_n$. This is indeed a homomorphism. To see if φ is injective, if we have $\varphi(a^k) = 0 \mod n$ then since $0 \le k \le n-1$, it must be that $k = 0$ since $a^k = 1$. To check surjectivity, we can see that $\forall k \in \mathbb{Z}_n, k = \varphi(a^k)$.

If $|G| = \infty$ then $G = \{\ldots, a^{-2}, a^{-1}, 1, a, a^2, \ldots\}$ with $a^k \neq 1\ \forall k \in \mathbb{Z}$. We now define $\varphi(a^k) = k$ and therefore $\varphi(a^k a^l) = \varphi(a^{k+l}) = k + l = \varphi(a^k) + \varphi(a^l)$. This proves it is a homomorphism. We can see that $\varphi(a^0) = 0$ so it is injective and $k = \varphi(a^k)$ for all $k \in \mathbb{Z}$ so it is surjective. □

This classifies all cyclic groups up to isomorphism. A natural question to ask from this is, can we classify all groups up to isomorphism?

no.

But if we modify the question, we can answer it and it has been answered.

Note, if $\varphi : G_1 \mapsto G_2$ is an injective homomorphism, it may not always be an isomorphism onto G_2 but it will always be an isomorphism onto its image

$$G_1 \cong \operatorname{im}\varphi$$

If G is cyclic, let $\langle a\rangle = G$, then to uniquely define a homomorphism, we only need to specify where to send a since $\varphi(a^k) = \varphi(a)^k$.

Theorem 2.10 (Cayley's Theorem)**.** *Every group is a group of permutations.*

Proof. Given any set A, there exists a symmetric group $S_A = \{\text{all bijections } \sigma : A \mapsto A\}$ where the operation is composition. Given G, consider S_G and given any $g \in G$, define $\sigma_g : G \mapsto G$ by $\sigma_g(x) = gx$. We claim that

1. Each σ_g is a permutation of the set A
2. $\varphi : G \mapsto S_G$ by $\varphi(g) = \sigma_g$ is a homomorphism
3. φ is injective

If we show this, then $G \cong$ a subgroup of S_G. Let us start by proving the first point. If $\sigma_g(x) = \sigma_g(y)$ then $gx = gy \to x = y$ and hence injective. Given anny $y =\in G$, $y = \sigma_g(g^{-1}y)$ which proves surjectivity and therefore σ_g is bijective and a permutation.

To prove claim 2, we consider $\varphi(gh) = \sigma_{gh}$ where $\sigma_{gh}(x) = ghx$. If we consider $\varphi(g)\varphi(h) = \sigma_g(\sigma_h(x)) = \sigma_g(hx) = ghx$ and therefore $\varphi(gh) = \varphi(g)\varphi(h)$. Let us note that $\varphi(g)$ is a permutation and the operation between $\varphi(g)$ and $\varphi(h)$ is function composition.

To prove the final claim, we just need see the kernel of φ. Let us suppose $\varphi(g) = \text{id}$ in S_G. Then $\sigma_g = \text{id}$ so $\forall x, gx = x \to g = e$ and therefore φ is injective. □

	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c		
c	c	b		

Table 3: Case 1 when $a^2 = 1$

2.2.1 Classifications of Small Groups

The trivial group is naturally isomorphic to $\mathbb{Z}_1$. We showed previously that groups of order 2 and 3 are fixed and therefore isomorphic to $\mathbb{Z}_2$ and $\mathbb{Z}_3$ respectively. For order 4, we need to split into cases. For the first case, we have that $a^2 = 1$. We can split this into further subcases. The first is if $b^2 = 1$ which then forces $c^2 = 1$ and therefore $bc = a$ and $cb = a$. This is a group. The second subcase is $b^2 \neq 1$ and so $b^2 = a$ which implies $cb = 1 = bc$ and $c^2 = a$. This is also a group.

In case 2, without a loss of generality (since we can just rename the elements), we let $a^2 \neq 1$ but rather $a^2 = b$. This gets the group table This also forms a group. If we look at this group and consider the mapping $1 \leftrightarrow 0, a \leftrightarrow 1, b \leftrightarrow$

	1	a	b	c
1	1	a	b	c
a	a	b	c	1
b	b	c	1	a
c	c	b	a	b

Table 4: Case 2 when $a^2 \neq 1$

$2, c \leftrightarrow 3$, we see that this is an isomorphism between $G \cong \mathbb{Z}_4$. If we look at the second subcase of case 1, we have the mapping $1 \leftrightarrow 0, b \leftrightarrow 1, a \leftrightarrow 2, c \leftrightarrow 3$ which is once again isomorphic to $\mathbb{Z}_4$. The first subcase is not isomorphic to $\mathbb{Z}$ since every element satisfies $x^2 = 1$. However, it is abelian and is called the *Klein 4-group.* It is not a cyclic group. It is denoted as V or K_4.

2.3 Back to the Symmetric Group

Let us consider $G = S_N$ and some cycle $(a_1, a_2, \ldots, a_k)$. We can write this cycle as a product of transpositions $(a_1, a_2, \ldots, a_k) = (a_1, a_k)(a_1, a_{k-1}) \ldots (a_1, a_2)$. Every cycle can be written as product of transpositions τ where $\tau^2 = 1$. Since every permutation is a product of cycles, we can deduce that every permutation is a product of cycles as well.

Theorem 2.11. *For any $\sigma \in S_n$, every way of writing it as a product of transpositions is either always even or odd.*

Proof. First let us show that the identity $1 = (a_1, b_1)(a_2, b_2) \ldots (a_m, b_m)$ must be such that m is even. We shall perform induction on m. We can see that $m \neq 1$ since $1 \neq (a, b)$. $m = 2$ is possible but 2 is even. So let us assume then that $m \geq 3$. By looking at the structure, we require that a_1 needs to show up again in a transposition to swap the a_1 back into place. Therefore, it must be true that some a_i/b_i. Without loss of generality, let us say $a_i = a_1$. Suppose that i is the first index for which this happens. Then we have

$$1 = (a_1, b_1) \ldots (a_{i-1}, b_{i-1})(a_i, b_i) \ldots (a_m, b_m)$$

If all the numbers are different, we can commute these transpositions. If either a_{i-1} or b_{i-1} is equal to b_i, we use the formula $(x, y)(x, z) = (x, z)(y, z)$ and we get that $(a_{i-1}, b_{i-1})(a_i, b_i) = (a_1, *)(*, *)$ where $* \neq a_1$. We can now move this transposition left without changing the number of transpositions or the number of a_1's in transpositions. Repeating this, we get

$$1 = (a_1, b_1)(a_1, c) \ldots (a_m, b_m)$$

We now have two cases. Case 1: $c = b_1$ and the two front terms cancel each other and then 1 can be written as a product of $m - 2$ transpositions and by induction, $m - 2$ is even and so is m.

Case 2: $c \neq b_1$ and therefore we have

$$\begin{aligned} 1 &= (a_1, b_1)(a_1, c)\dots(a_m, b_m) \\ &= (a_1, c)(b_1, c)\dots(a_m, b_m) \end{aligned}$$

which means we have one fewer a_1 than with what we started with. Therefore there exists another a_1 somewhere and we have to keep doing this process. If we get lucky, we go back to case 1 and the number of products is even. If we're not lucky, we get

$$1 = (a_1, *)\dots(*, *)$$

where again, $* \neq 1$. This is impossible. Therefore the number of transpositions must be even in order to recover the identity. Finally, if σ is any permutation, we can write it as $\sigma = \tau_a \dots \tau_m$ or as $\sigma = \tau_1' \dots \tau_l'$ since a permutation can be expressed with different products of transpositions. Then we have

$$\tau_m \dots \tau_1 \tau_1' \dots \tau_l' = 1$$

Since there are $m + l$ transpositions and by the previous case we showed that there must be an even number of transpositions, $m + l$ must be even and therefore m, l must have the same parity. □

Definition 2.12. We say $\sigma \in S_n$ is even if it can be written as an even product of transpositions and is odd otherwise.

The sign of a permutation, denoted by $\text{sign}(\sigma)$, is

$$\text{sign}(\sigma) = (-1)^{\text{num of transpositions } \sigma \text{ can be written as prodct}} = \begin{cases} 1 & \sigma \text{ even} \\ -1 & \sigma \text{ odd} \end{cases}$$

Proposition 2.13. *Half of the elements of S_n are even and half are odd*

Proof. Fix any transposition $\tau \in S_n$. Consider the map even permutations$\mapsto$odd permutations given by $\sigma \mapsto \tau\sigma$. We claim that this is a bijection. If $\tau\sigma = \tau\sigma' \to \sigma = \sigma'$ and therefore it is one to one. For any odd permutation ρ, then there exists $\tau(\tau\rho)$ where $\tau\rho$ is an even permutation. Therefore it is also onto. □

We note that the set of even permutations in S_n is a subgroup. It is known as the alternating group on n letters and we denote it as A_n. $|A_n| = \frac{n!}{2}$. For $n \geq 4$, A_n is non abelian.

3 Structure of Groups

Definition 3.1. An element of a group G is said to have order n if $g^n = 1$ and $g^k \neq k$ for every k such that $0 < k < n$. If $g^n \neq 1$ for any n, then g has infinite order. The order of g is also the order $\langle g \rangle$.

As a consequence, g generates a cyclic group G if and only if the order of g is $|G|$.

Proposition 3.2. *The order of g divides $|G|$*

Proof. If $|G| = \infty$, there is nothing to show. If the order of g is k, then $1, g, g^2, \dots, g^{k-1}$ are k distinct elements. If this is all of g, we're done. If not, then for some $a_1 \in G$ that isn't above, you can obtain k more elements in g. We can do this until all of our remaining elements have been saturated. As such, we get that $|G| = nk$. □

We don't have to worry about $a_1 a_2$ since suppose $a_i g^k = a_j g^l$ and without a loss of generality with $j > 1$, then $a_j = a_i g^{k-l}$ so a_j would have appeared earlier.

We can show that $\mathbb{Z} \times \mathbb{Z} \ncong \mathbb{Z}$. Suppose there exists (a, b) such that it generates all of $\mathbb{Z}^2$. Then $n(a, b) = (1, 0)$ for some integer n which forces $b = 0$. We can use this same logic to say there exists an integer m such that $m(a, b) = (0, 1)$ which forces $a = 0$ but $(0, 0)$ can't generate $\mathbb{Z}^2$ therefore $\mathbb{Z}^2$ cannot be cyclic.

Proposition 3.3. *$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ if and only if m and n are coprime.*

Proof. ($\Leftarrow$) We assume that $\gcd(m,n)=1$. We will show that $(1,1)$ has order mn in $\mathbb{Z}_m \times Z_n$. Clearly, $mn(1,1)=0$ in $\mathbb{Z}_m \times Z_n$. However, there might be an integer smaller than mn. We ask if there exists a k such that $k(1,1)=(0,0)$ for some k such that $0<k<mmn$. Then we must have that $k(1,1)=(k \mod m, k \mod n)$ and if this is 0, then $m|k$ and $n|k$. Since they're coprime, $mn|k$ which means that $mn \leq k$ so our previous assumption is wrong and $(1,1)$ has order mn in $\mathbb{Z}_m \times \mathbb{Z}_n$ and hence is isomorphic to Z_{mn}.

($\Rightarrow$) Now suppose that their gcd isn't one. Let $a>1$ such that $a|m$ and $a|n$. Take any $(x,y) \in Z_m \times \mathbb{Z}_n$, then $\frac{mn}{a}(x,y)=(m\frac{n}{a} \mod m, n\frac{m}{a} \mod n)=(0,0)$. This means that every element in $\mathbb{Z}_m \times Z_n$ has order $\leq \frac{mn}{a} \leq mn$ and thus $\mathbb{Z}_m \times \mathbb{Z}_n$ cannot be cyclic. □

Definition 3.4. A group is *finitely generated* if it can be generated by finitely many elements

Proposition 3.5. $\mathbb{Q}$ *is not finitely generated*

Proof. Suppose we can generate $\mathbb{Q}$ by $\frac{a_1}{b_1}, \ldots, \frac{a_n}{b_n}$. Let p be any prime $> b_1 b_2 \ldots b_n$. Then $\frac{1}{p} \in \mathbb{Q} \neq m_1\frac{a_1}{b_1}+\ldots+m_n\frac{a_n}{b_n}$ for all $m_1, \ldots, m_n$. □

Proposition 3.6. $\mathbb{R}$ *is not finitely generated*

Proof. Suppose $x_1, \ldots, x_n$ generates $\mathbb{R}$. The set $\{m_1x_1+\ldots+m_nx_n : m_1, \ldots, m_n \in \mathbb{Z}\}$ is countable but $\mathbb{R}$ is not. □

Theorem 3.7 (Fundamental Theorem of Finitely Generated Abelian Groups)**.** *Every finitely generated abelian group is isomorphic to* $\mathbb{Z}_{p_1^{k_1}} \times \mathbb{Z}_{p_2^{k_2}} \times \ldots \times Z_{p_n^{k_n}} \times \mathbb{Z}^n$ *where* $p_1, \ldots, p_n$ *are primes (where we can have repeats) and* $n, p_1, \ldots, p_n, k_1, \ldots, k_n$ *are uniquely determined.*

Corollary 3.8. *Every finite abelian group is isomorphic to* $\mathbb{Z}_{p_1^{k_1}} \times \mathbb{Z}_{p_2^{k_2}} \times \ldots \times Z_{p_n^{k_n}}$

3.1 Cosets

Definition 3.9. If G is a group and $H \leq G$ then for $a \in G$, we define the *left coset* of H in G and we represent it to be

$$aH := \{ah : h \in H\}$$

Proposition 3.10. *Two cosets are either disjoint or equal and* $G = \bigcup_{a \in G} aH$

Proof. Suppose $aH \cap bH \neq \emptyset$. Since the intersection isn't empty, there exists $ah_1 = bh_2$ for some $h_1, h_2 \in H$. Now let $ah \in aH$ be any element, then $ah = bh_2h_1^{-1}h \in bH$ since H is a subgroup and therefore closed under group multiplication. We therefore have the $aH \subseteq bH$. Similarly, $\forall bh' \in bH, bh' = ah_1h_2h^{-1} \in aH$ and we have $bH \subseteq aH$ and therefore $aH = bH$.

$\bigcup aH \subseteq G \ \forall a \in G$ and $a \in aH$ (since $1 \in H$) and so $G \subseteq \bigcup aH$ and therefore $\bigcup aH = G$. □

For every coset, $|aH| = |H|$ since multiplying by a is a bijection. Since $G = \bigcup_{\text{distinctcosets}} aH$, the order of $|G| = (\ \#$ distinct cosets of $H)|H|$. This is effectively the proof for Lagrange's Theorem.

Theorem 3.11 (Lagrange's Theorem)**.** *If* G *is a finite group, the order of any subgroup of* G *divides the order of* G*.*

Corollary 3.12. *In any finite group* G*, the order of an element must divide* $|G|$

Proof. Ask a toddler on the street. □

Corollary 3.13. *Any group of prime order of* p *is isomorphic to* $\mathbb{Z}_p$*.*

Proof. Suppose $|G| = p$. Take any nontrivial element x. Then $\langle x \rangle$ is nontrivial so by Lagrange's theorem, it must have order p. So $\langle x \rangle = G$ and thus G is cyclic and $G \cong \mathbb{Z}_p$. □

Definition 3.14. The number of distinct cosets of H in G is called the index of H in G and denoted as $(G:H)$.

When G is finite, then $(G:H) = \frac{|G|}{|H|}$.

Proposition 3.15. $aH = bH$ *if and only if* $b^{-1}a \in H$

Proof. ($\Rightarrow$) Suppose $aH = bH$. Since $a \in aH$, that means $a = bh$ for some $h \in H$ and then we have $b^{-1}a = h$.

($\Leftarrow$) Suppose $b^{-1}a \in H$. For every $ah \in aH$, then we have that $ah = bb^{-1}ah \in bH$ since $b^{-1}a$ and h are both in h and therefore $aH \subseteq bH$. Because cosets must be disjoint or equivalent, $aH = bH$. □

We can effectively define an equivalence relation $\sim$ by $a \sim b$ if $b^{-1}a \in H$. The equivalence classes of this relation are the left cosets of H in G. We can also have the equivalence condition for right cosets defined as

$$H_a = \{ha : h \in H\}$$

to be $ab^{-1} \in H$.

3.2 Normal Subgroups

Definition 3.16. A subgroup $H \in G$ is called *normal* if $\forall g \in G$ and $h \in H$, $ghg^{-1} \in H$ or $gHg^{-1} \subseteq H \ \forall g \in G$. We denote it

$$H \lhd G$$

Trivially, every subgroup of an Abelian group is normal. We call ghg^{-1} the *conjugation* of h by g.

Theorem 3.17. *For a subgroup H of G, the following are equivalent:*

1. $H \lhd G$
2. $aH = Ha \ \forall a \in G$
3. *Every left coset of H in G is a right coset and conversely*
4. $\forall a, b \in G$ *and* $h_1, h_2 \in H$, $ah_1bh_2 \in abH$

Proof. Suppose $H \lhd G$. For every $h \in H$, $ah = aha^{-1}a \in H$ and we can show the same for the other inclusion or use that cosets must be disjoint or equivalent. Two implies three is trivial. For the three implies two, suppose that $aH = Hb$ for some b. We have that $a \in aH$ an by our assumption, $a = h_1b$ for some b. Simlarly, $b \in Hb$ so $b = ah_2$ for some $h_2 \in H$. Suppose $h'b \in Hb \ \forall h' \in H$, then $h'b = h'h_1^{-1}a \in Ha$ and therefore $Hb \cap Ha \neq \emptyset$ and therefore they must be the same. For two implies four, we have that $ah_1bh_2 = abb^{-1}h_1bh_2$ where $b^{-1}h_1b \in H$ by $bH = Hb \to bh_1 = h'b \to bh_1b^{-1} = h'$ and therefore $ah_1bh_2 \in abH$. Finally, we show that four implies one since $aha^{-1} = aha^{-1}e \in aa^{-1}H$ □

Now suppose that $N \lhd G$. Let G/N (said as "G mod N") be the set of left cosets of N in G and define an operation on this bet $(aN)(bN) = abN$. We should note that whenever we are working with equivalence classes, we require that functions on them must be *well defined* which means it does not depend on the choice of the representative of the equivalence class. Let us take fractions for example. If we define $f(\frac{m}{n}) = m^2$ we see that it is not well defined because $f(\frac{1}{2}) \neq f(\frac{2}{4})$. If instead we had $f(\frac{m}{n}) = \frac{m^2}{n^2}$, this would be well defined.

Theorem 3.18. *G/N is a group defined as $G/N := \{aN : a \in G\}$ with its group operation being $(aN)(bN) = abN$. This is known as the quotient group of G mod N.*

Proof. We first need to show that the operation is well defined. If $aN = cN$ and $bN = dN$, we need that $abN = cdN$. If $aN = cN$, therefore $a^{-1}c \in N$ and if $bN = dN$ then $b^{-1}d \in N$. Then we can say $c = an_1$ and $d = bn_2$ for some $n_1, n_2 \in N$ and hence $cd = an_1bn_2 \in abN$ since N is normal. Since $cdN \cap abN \neq \emptyset$, $cdN = abN$. We now check group axioms. For associativity, we have $(aNbN)cN = abNcN = abcN = a(bc)N = aN(bNcN)$. For the identity, we have N since $aNN = aN = N(aN)$. For inverses, we have that $(aN)^{-1} = a^{-1}N$ since $aNa^{-1}N = N$. □

We can say that $\mathbb{Z}/3\mathbb{Z}$ is generated $1 + n\mathbb{Z}$ and is isomorphic to Z_n.

Proposition 3.19. *For a homomorphism $\varphi : G \to H$, then* $\ker \varphi \lhd G$

Proof. Take some $k \in \ker \varphi$ and $g \in G$. Then $\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) = \varphi(g)e_H\varphi(g)^{-1} = e_H$. Therefore $gkg^{-1} \in \ker \varphi$. □

Theorem 3.20 (First Isomorphism Theorem)**.** *If $\varphi : G \mapsto H$ is any homomorphism, then* $\operatorname{im} \varphi \cong G/\ker \varphi$.

Proof. Define $\Phi : G/\ker\varphi \mapsto \operatorname{im}\varphi$ by $\Phi(a\ker\varphi) = \varphi(a)$. We first need to show that Φ is well defined, a homomorphism, and is bijective. First, if $a\ker\varphi = b\ker\varphi$ then $a^{-1}b \in \ker\varphi$. We then have that $b = ak$ for some $k \in \ker\varphi$. So $\Phi(b\ker\varphi) = \varphi(b) = \varphi(ak) = \varphi(a)\varphi(k) = \Phi(a\ker\varphi)$. To show it is a homomorphism, we have that $\Phi((a\ker\varphi)(b\ker\varphi)) = \Phi(ab\ker\varphi) = \varphi(ab) = \varphi(a)\varphi(b) = \Phi(a\ker\varphi)\Phi(b\ker\varphi) = \Phi((a\ker\varphi)(b\ker\varphi))$. Φ is surjective because for any $y \in \operatorname{im}\varphi$, $y = \varphi(x)$ for some $x \in G$ so $y = \Phi(x\ker\varphi)$. Φ is injective since $\Phi(a\ker\varphi) = 1 \to \varphi(a) = 1 \therefore a \in \ker\varphi$. Therefore $a\ker\varphi = \ker\varphi$ and $a = e$. □

Definition 3.21. A *fiber* of a map $f : X \to Y$ is an equivalence class on X for having the same image in Y under f. If y is the image, then we can define this equivalence class to be $f^{-1}(y) := \{x \in X : f(x) = y\}$.

What the First Isomorphism Theorem is saying is to take the fibers of φ on H and collapse them all to one point that represents each equivalence class.

Proposition 3.22. *$\mathbb{R}/\mathbb{Z} \cong S^1$ where S^1 is the unit circle defined as $S^1 := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$.*

Proof. In polar form, we can define the unit circle as $\{e^{i\theta} : \theta \in [0, 2\pi)\}$. Let us suppose the homomorphism $\varphi : \mathbb{R} \to S^1$ by $\varphi(x) = e^{2\pi ix}$. This is trivially a homomorphism. The kernel is just $\mathbb{Z}$ and the image is all of S^1. Therefore by the First Isomorphism Theorem, $G/\mathbb{Z} \cong S^1$. □

Example 3.23. If we define the homomorphism $\varphi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ by the map $\varphi(a, b) = a + b$, then the kernel is $\ker\varphi = \{(a, -a) : a \in \mathbb{Z}\} = \langle(1, -1)\rangle$. The image is just the integers so $\mathbb{Z} \times \mathbb{Z}/\langle(1, -1)\rangle \cong \mathbb{Z}$.

3.3 Centers and Commutators

Definition 3.24. The *center* of G is defined to be

$$Z(G) := \{g \in G : ag = ga \ \ \forall a \in G\}$$

Proposition 3.25.

$$Z(D_n) = \begin{cases} \{1, \rho^{\frac{n}{2}}\} & n \text{ even} \\ \{1\} & \text{otherwise} \end{cases}$$

Proof. Assume $n = 2k$. We use the fact that $\mu\rho^k = \rho^{-k}\mu \Rightarrow \mu\rho^k = \rho^k\mu$ if $n = 2k$. This means that ρ commutes with μ. ρ^k also commutes with any ρ^l for $l = 0, 1, \ldots, n-1$ just by commutativity of addition. ρ^k also commutes with $\mu\rho^l$ due to the above facts and therefore $\{1, \rho^k\} \subseteq Z(D_{2k})$. We note that ρ^l for any $l \neq 0, k$ cannot commute since $\mu\rho^l = \rho^{-l}\mu = \rho^{n-l}\mu$ and $\rho(\mu\rho^l) = \mu\rho^{l-1}$ whereas $\mu\rho^l\rho = \mu\rho^{l+1}$ and so we've exhausted all other possibilities and hence the center is just $Z(D_{2k}) = \{1, \rho^{\frac{n}{2}}\}$. For odd, we apply the same logic above and so the center is just the trivial element $Z(D_{2k+1}) = \{1\}$. □

Definition 3.26. The commutator subgroup of G is defined to be

$$G' := \langle\{aba^{-1}b^{-1} : a, b \in G\}\rangle$$

Proposition 3.27. *1. $G' \lhd G$*

2. G/G' is Abelian

3. For $N \lhd G$, G/N is abelian if and only if $G' \subseteq N$.

Proof. 1. $g(aba^{-1}b^{-1})g = (gag^{-1})(gbg^{-1})(ga^{-1}g^{-1})(gb^{-1}g^{-1}) = xyx^{-1}y^{-1} \in G'$ where $x = gag^{-1}$ and $y = gbg^{-1}$. This is true for every $g \in G$ and every generator $aba^{-1}b^{-1}$ of G' and hence $G' \lhd G$.

2. Take $aG', bG' \in G/G'$, then $aG'bG' = abG' = abb^{-1}a^{-1}baG' = baG' = bG'aG'$ since $b^{-1}a^{-1}ba \in G'$ as it is a commutator. Alternatively we can say that $abG' = baG' \Leftrightarrow (ba)^{-1}ab = a^{-1}b^{-1}ab \in G'$.

3. Suppose $G' \subseteq N$, then G/N is abelian. $aNbN = abN = baN = bNaN$ where commutativity comes from the fact that $(ba)^{-1}ab \in G' \subseteq N$. Now we suppose that G/N is abelian, then $(ba)^{-1}ab \in N \ \ \forall b, a \in G$ so $G' \subseteq N$ since $abN = baN \to N = (ba)^{-1}abN$.

□

Definition 3.28. A group is *simple* if it has no proper nontrivial normal subgroups.

Theorem 3.29. *A_n is simple for all $n \geq 5$.*

3.4 Group Actions and Orbits

Definition 3.30. Suppose G is any group and X is a set. We say G *acts* on X (or X is a G-set) if to any $g \in G, x \in X$, we assign a point $g \cdot x \in X$ satisfying

1. $e \cdot x = x \ \forall x \in X$
2. $h \cdot (g \cdot x) = (hg) \cdot x$

Example 3.31. $\mathbb{Z}$ is a group action on the set $\mathbb{R}$ by a translation $n \cdot x = n + x$. Additionally, $\mathbb{R}$ is a group on $\mathbb{R}^2$ by $\theta \in \mathbb{R}, \vec{x} \in \mathbb{R}^2$, $\theta \cdot \vec{x} = \vec{x}$ rotated by θ radians counter clockwise.

Definition 3.32. Suppose G acts on X. Given $x \in X$, we define

$$G_x := \{g \in G : g \cdot x = x\}$$

This is known as the *isotropy subgroup* or *stabilizer* of x in G.

Proposition 3.33. *The stabilizer of x in G is a subgroup.*

Proof. $1 \in G_x$ since 1 fixed everything. Suppose $g \cdot x = x$ and $h \cdot x = x$, then $(gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = x$. Now suppose that $g \in G_x$, then that implies $g \cdot x = x \to g^{-1}g \cdot x = g^{-1} \cdot x \to 1 \cdot x = g^{-1} \cdot x$ and therefore $g^{-1} \in G_x$. □

Definition 3.34. Fix $x \in X$. The *orbit* of G on x is

$$Gx := \{g \cdot x : g \in G\} \subseteq X$$

Proposition 3.35. *Orbits are disjoint.*

Proof. Suppose $Gx \cap Gy \neq \emptyset$. Then $g \cdot x = h \cdot y$ for some $g, h \in G$. Then $y = h^{-1}g \cdot x$ and $x = g^{-1}h \cdot y$. For some $a \cdot x \in Gx$, then we have $a \cdot x = ag^{-1}h \cdot y \in Gy$ and likewise for some $b \cdot y \in Gy$, then $by = bh^{-1}g \cdot x \in Gx$ and hence we $Gy \subseteq Gx$ and $Gx \subseteq Gy$. □